

() Preliminary Specification
(V) Final Specification

Module	19" SXGA TFT-LCD Module
Model Name	G190ETN01.0 (002)

Customer

Date

Approved by

Note: This Specification is subject to change without notice.

Checked & Approved by

Date

Elsie.MK.Kuo

2023/01/13

Prepared by

Date

Daniel Chang

2023/01/13

GD Display Business Unit / AU Optronics corporation

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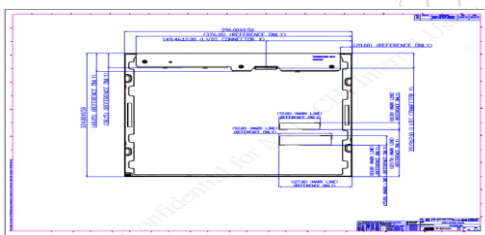
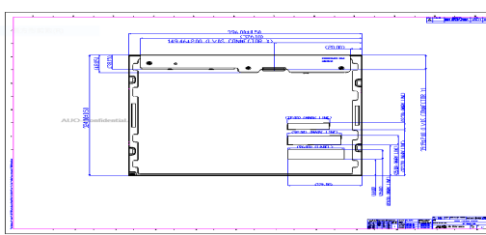
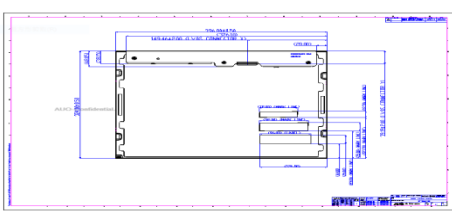
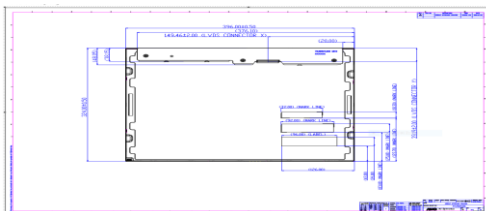
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1.2.1 2022/12/12

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7.2 Backlight Unit

Physical interface is described as for the connector on module. These connectors are capable of accommodating the following signals and will be following components.

7.2.1 Connector

Connector Name / Designation	Light Bar Connector
Manufacturer	ENTERY INDUSTRIAL CO., LTD
Type Part Number	3707K-S06N-21R

7.2.1 Connector

Connector Name/ Designation	Light Bar Connector
Manufacturer	CvLux
Type Part Number	C11406M1HRN-NH1
Mating Housing Manufacturer	CvLux or compatible
Mating Housing Part Number	C11406S0000-NH (non-lock type) C11406SL000-NH (lock type) or compatible

1.2.2 2023/01/13

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7.2.1 Connector

Connector Name/ Designation	Light Bar Connector
Manufacturer	CvLux
Type Part Number	C11406M1HRN-NH1
Mating Housing Manufacturer	CvLux or compatible
Mating Housing Part Number	C11406S0000-NH (non-lock type) C11406SL000-NH (lock type) or compatible

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Mating Housing Manufacturer	CvLux or compatible
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1. Handling Precautions

- 1) Since front polarizer is easily damaged, please be cautious and not to scratch it.
- 2) Be sure to turn off power supply when inserting or disconnecting from input connector.
- 3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or soft cloth.
- 5) Since the panel is made of glass, it may be broken or cracked if dropped or bumped on hard surface.
- 6) To avoid ESD (Electro Static Discharge) damage, be sure to ground yourself before handling TFT-LCD Module.
- 7) Do not open nor modify the module assembly.
- 8) Do not press the reflector sheet at the back of the module to any direction.
- 9) In case if a module has to be put back into the packing container slot after it was taken out from the container, do not press the center of the LED light bar edge. Instead, press at the far ends of the LED light bar edge softly. Otherwise the TFT Module may be damaged.
- 10) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
- 11) TFT-LCD Module is not allowed to be twisted & bent even force is added on module in a very short time. Please design your display product well to avoid external force applying to module by end-user directly.
- 12) Small amount of materials without flammability grade are used in the TFT-LCD module. The TFT-LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
- 13) Severe temperature condition may result in different luminance, response time and lamp ignition voltage.
- 14) Continuous operating TFT-LCD display under low temperature environment may accelerate lamp exhaustion and reduce luminance dramatically.
- 15) The data on this specification sheet is applicable when LCD module is placed in landscape position.
- 16) Continuous displaying fixed pattern may induce image sticking. It's recommended to use screen saver or shuffle content periodically if fixed pattern is displayed on the screen.

2. General Description

G190ETN01.0 is a Color Active Matrix Liquid Crystal Display composed of a TFT-LCD panel, a driver circuit, and a backlight system. The screen format is intended to support the SXGA (1280(H) x 1024(V)) screen and 16.7M colors (RGB 6-bits + HiFRC data). All input signals are 2-channel LVDS interface.

2.1 Display Characteristics

The following items are characteristics summary on the table under 25 °C condition:

Items	Unit	Specifications
Screen Diagonal	[mm]	482.6 (19.0")
Active Area	[mm]	376.32 (H) x 301.06 (V)
Pixels H x V		1280(x3) x 1024
Pixel Pitch	[mm]	0.294 (per one triad) x 0.294
Pixel Arrangement		R.G.B. Vertical Stripe
Display Mode		Normally White
White Luminance	[cd/m²]	350 (center, Typ) @60.0mA
Contrast Ratio		1000 : 1 (Typ)
Optical ResponseTime	[msec]	5 ms(Typ, on/off)
Nominal Input Voltage VDD	[Volt]	+5.0 V
Power Consumption	[Watt]	13.9 W (Typ) (PDD= 4.7 W, PLED=9.2 W)
Weight	[Grams]	1800 (Typ)
Physical Size (H x V x D)	[mm]	396 (H) x 324 (V) x 11.2(D) (Typ)
Electrical Interface		Dual channel LVDS
Surface Treatment		Anti-Glare
Support Color		16.7M colors (RGB 6-bit + Hi_FRC)
Temperature Range Operating Storage (Non-Operating)	[°C] [°C]	0 to +50 -20 to +60
RoHS Compliance		RoHS Compliance

2.2 Optical Characteristics

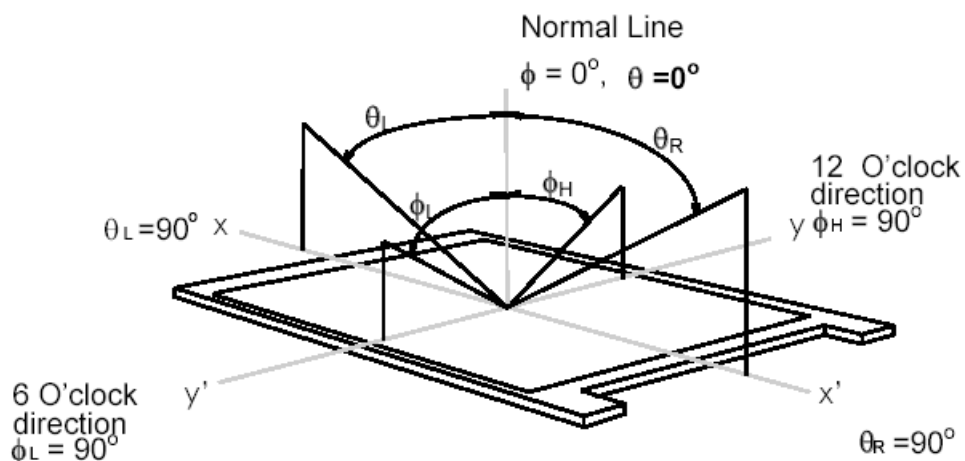
The optical characteristics are measured under stable conditions at 25°C (Room Temperature).

Item	Unit	Conditions	Min.	Typ.	Max.	Note
Viewing Angle	[degree]	Horizontal (Right) CR = 10 (Left)	75 75	85 85	-	2
		Vertical (Up) CR = 10 (Down)	70 70	80 80	-	
Contrast Ratio		Normal Direction	600	1000	-	3
Optical Response Time	[msec]	Raising Time (TrR)	-	3.6	5.7	4
		Falling Time (TrF)	-	1.4	2.3	
		Rising + Falling	-	5	8	
Color / Chromaticity Coordinates (CIE)		Red x	0.629	0.679	0.729	5
		Red y	0.263	0.313	0.363	
		Green x	0.235	0.285	0.335	
		Green y	0.609	0.659	0.709	
		Blue x	0.099	0.149	0.199	
		Blue y	0.012	0.062	0.112	
		White x	0.263	0.313	0.363	
		White y	0.279	0.329	0.379	
Central Luminance	[cd/m ²]		280	350	-	6
Luminance Uniformity	[%]	9 Points	75	80	-	7
NTSC	[%]			85		

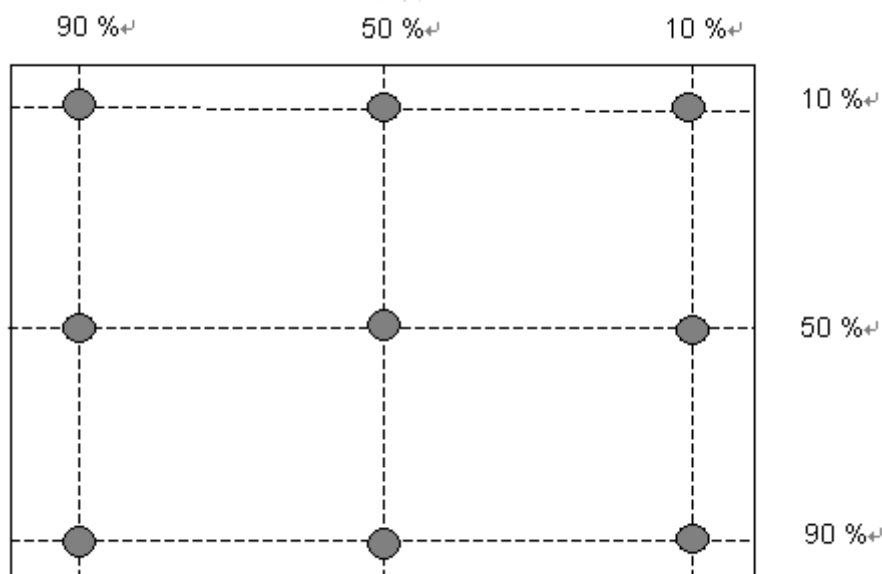
Optical Equipment: BM-5A, BM-7, PR880, or equivalent

Note 1: Definition of viewing angle

Viewing angle is the measurement of contrast ratio ≥ 10 , or ≥ 5 , at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as follows; 90° (θ) horizontal left and right and 90° (Φ) vertical, high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated about its center to develop the desired measurement viewing angle.



Note 2: 9 points position

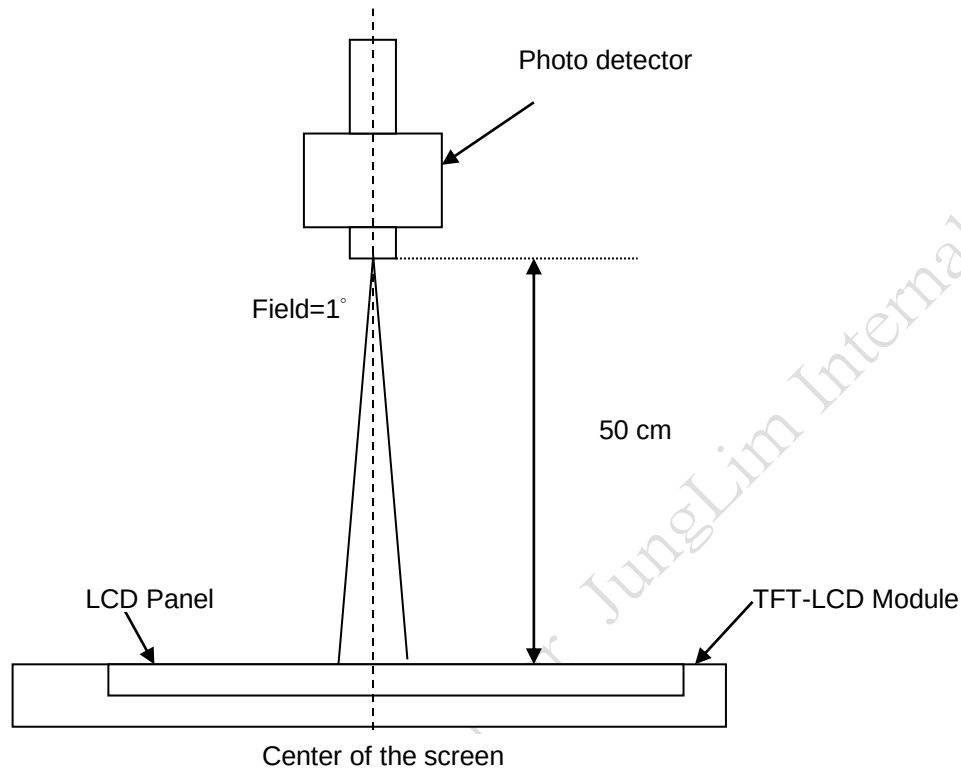


Note 3:

$$\text{Uniformity} = \frac{\text{Minimum Luminance in 9 points(1-9)}}{\text{Maximum Luminance in 9 Points(1-9)}}$$

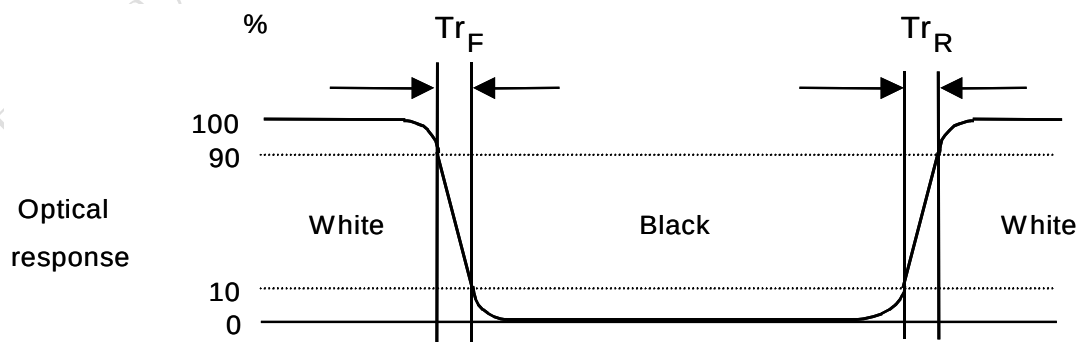
Note 4: Measurement method

The LCD module should be stabilized at given temperature for 30 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 30 minutes in a stable, windless and dark room.



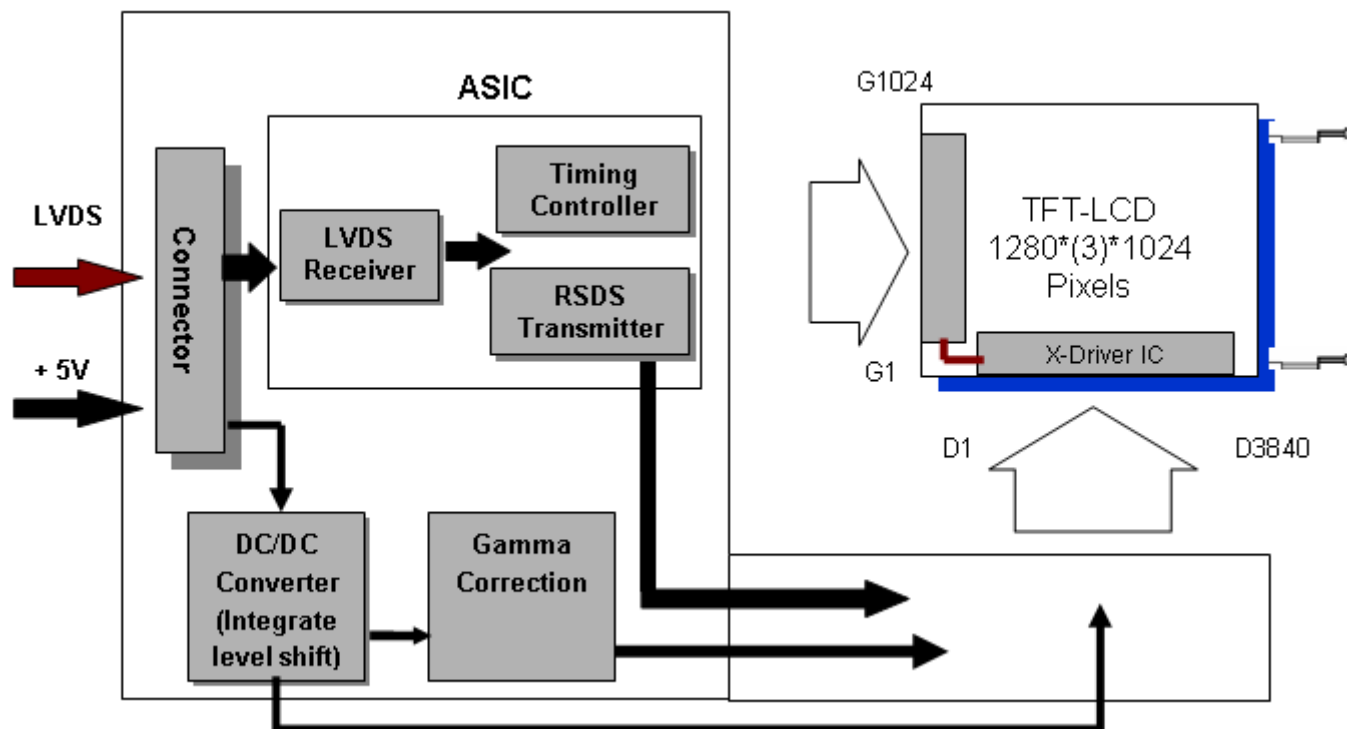
Note 5: Definition of response time:

The output signals of photo detector are measured when the input signals are changed from “Full Black” to “Full White” (rising time), and from “Full White” to “Full Black” (falling time), respectively. The response time is interval between the 10% and 90% of amplitudes. Please refer to the figure as below.



3. Functional Block Diagram

The following diagram shows the functional block of the 19.0 inches wide Color TFT-LCD Module:



4. Absolute Maximum Ratings

Absolute maximum ratings of the module are as following:

4.1 TFT LCD Module

Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	VDD	-0.3	+6.0	[Volt]	Note 1,2

4.2 Absolute Ratings of Environment

Item	Symbol	Min.	Max.	Unit	Conditions
Operating Temperature	TOP	0	+50	[°C]	Note 3
Glass surface temperature (operation)	TGS	0	+65	[°C]	Note 3, Note 4
Operation Humidity	HOP	5	90	[%RH]	Note 3
Storage Temperature	TST	-20	+60	[°C]	
Storage Humidity	HST	5	90	[%RH]	

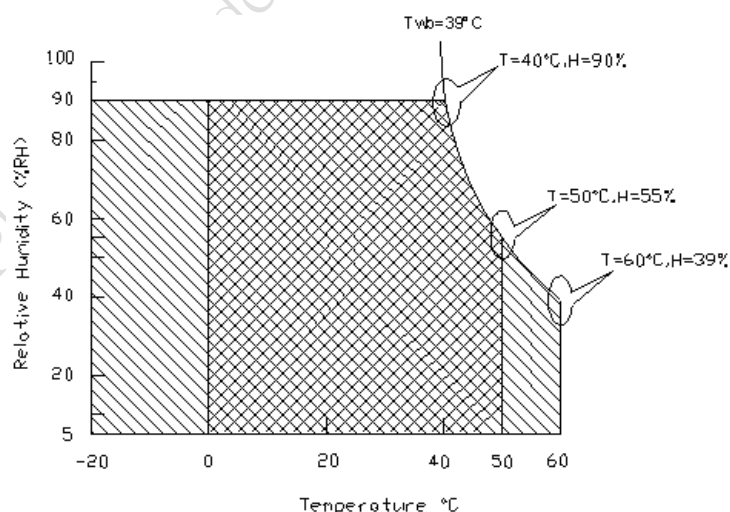
Note 1: With in Ta (25°C)

Note 2: Permanent damage to the device may occur if exceeding maximum values

Note 3: Temperature and relative humidity range are shown as the below figure.

1. 90% RH Max (Ta ≤ 39°C)
2. Max wet-bulb temperature at 39°C or less. (Ta ≤ 39°C)
3. No condensation

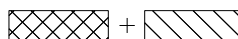
Note 4: Function Judged only



Operating Range



Storage Range



5. Electrical characteristics

5.1 TFT LCD Module

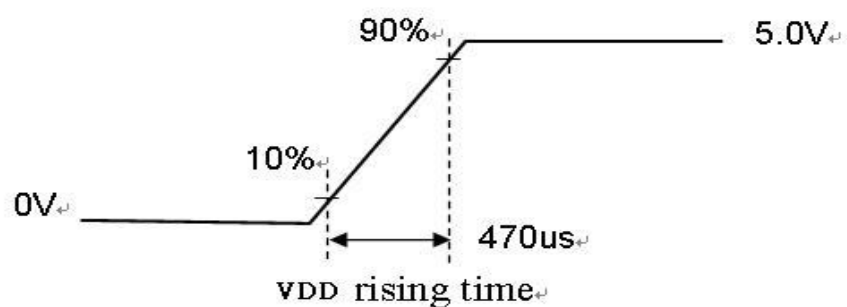
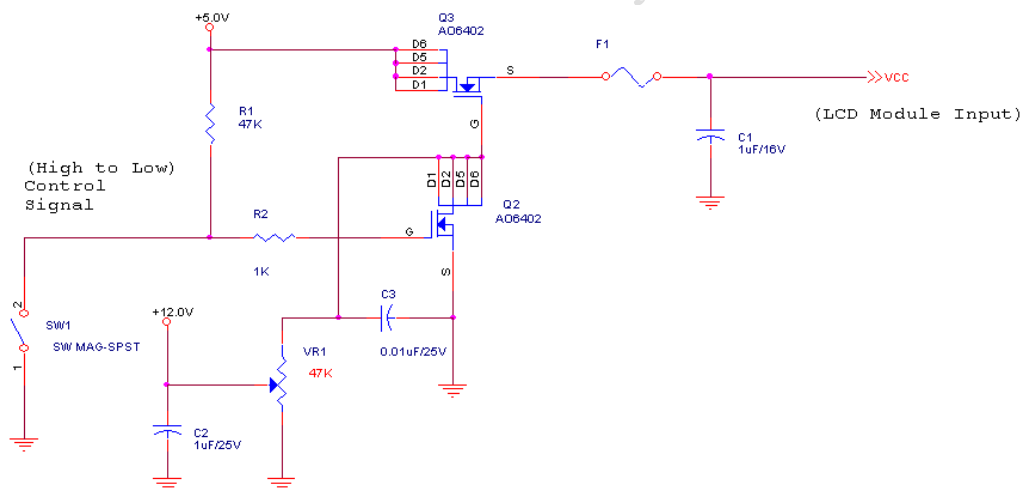
5.1.1 Power Specification

Input power specifications are as follows:

Symbol	Parameter	Min	Typ	Max	Unit	Conditions
VDD	Logic/LCD Drive Voltage	4.5	5.0	5.5	[Volt]	+/-10%
IDD	Input Current	-	0.94	1.1	[A]	VDD= 5.0V, All Black Pattern At 60Hz,
PDD	VDD Power	-	4.7	5.5	[Watt]	VDD= 5.0V, All Black Pattern At 60Hz
IRush	Inrush Current	-	-	3.0	[A]	Note 1
VDDrp	Allowable Logic/LCD Drive Ripple Voltage	-	-	100	[mV] p-p	VDD= 5.0V, All Black Pattern At 75Hz

Note 1: Measurement conditions:

The duration of rising time of power input is 470us.



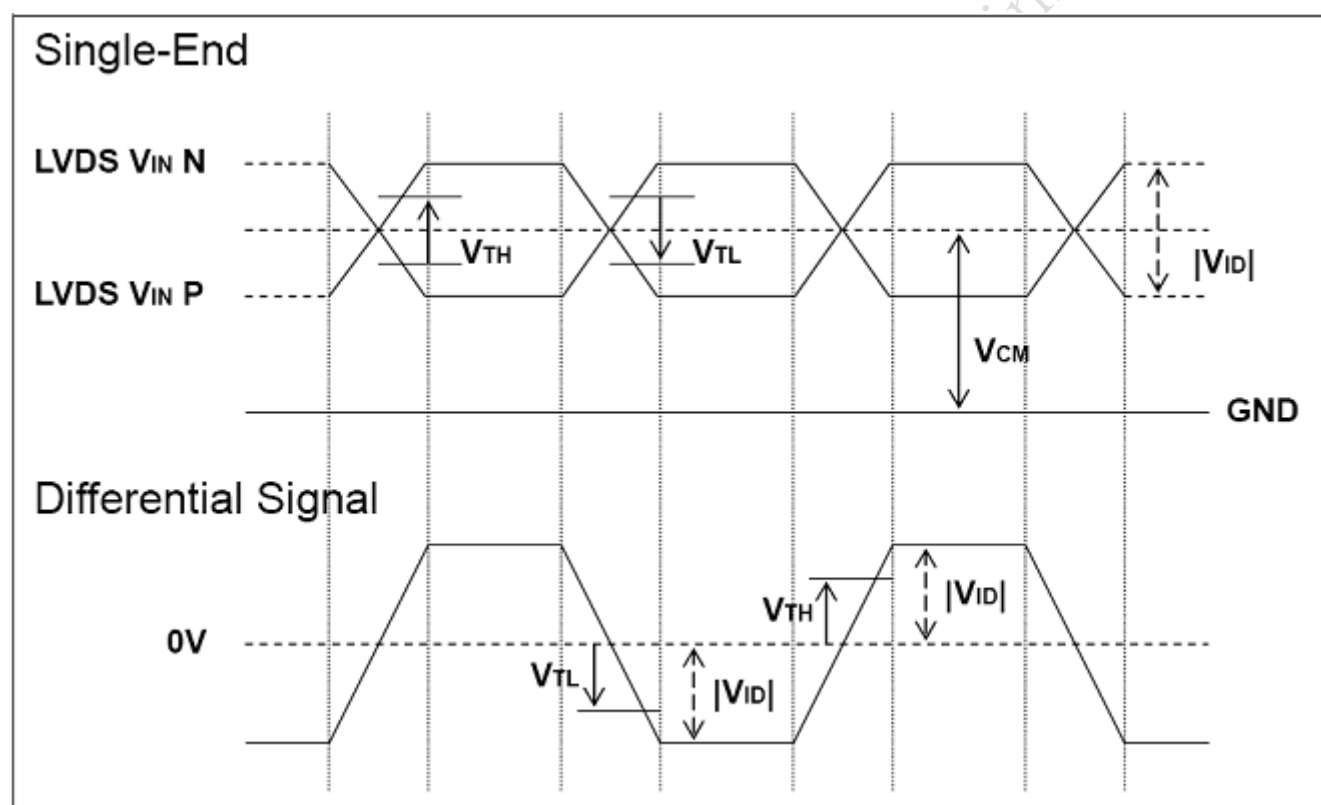
5.1.2 Signal Electrical Characteristics

Input signals shall be low or Hi-Z state when VDD is off. Please refer to specifications of SN75LVDS82DGG (Texas Instruments) in detail.

Each signal characteristics are as follows;

Symbol	Parameter	Min	Typ	Max	Units	Condition
V_{TH}	Differential Input High Threshold	-	-	+100	[mV]	$V_{CM} = 1.2V$ Note 1
V_{TL}	Differential Input Low Threshold	-100	-	-	[mV]	$V_{CM} = 1.2V$ Note 1
$ V_{ID} $	Input Differential Voltage	100	400	600	[mV]	Note 1
V_{CM}	Differential Input Common Mode Voltage	+1.0	+1.2	+1.5	[V]	$V_{TH}-V_{TL} = 200mV$ (max) Note 1

Note1: LVDS Signal Waveform



5.1.3 Backlight unit

Parameter guideline for LED driving is under stable conditions at 25℃ (Room Temperature):

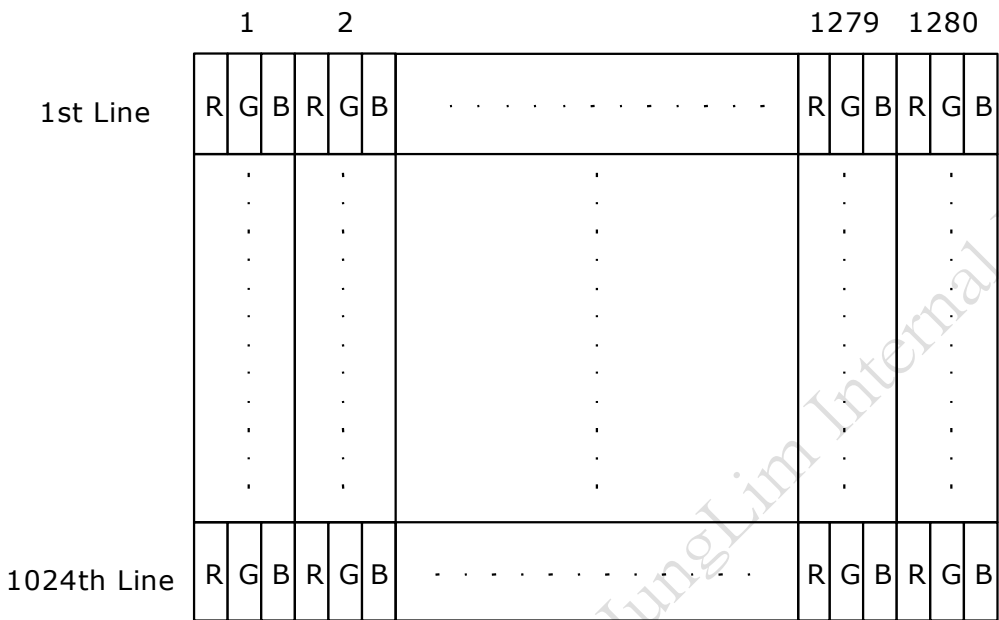
Symbol	Parameter	Min.	Typ.	Max.	Unit	Note
IR _{LED}	LED Operation Current	-	60	-	[mA] Note 1	Operating with fixed driving current
V _{LB}	Light Bar Operation Voltage (for reference)	-	38.4	41.6	[Volt] Note 2	
P _{BLU}	BLU Power consumption (for reference)	-	9.2	10.3	[Watt]	
LT _{LED}	LED life Time (Typical)	30,000		-	[Hour] Note 3	

- Note 1 :The specified current is input LED chip 100% duty current.
- Note 2 : The value showed in the table is one light bar’s operation voltage.
- Note 3 : Definition of life time : brightness becomes 50% of its original value. The minimum life time of LED unit is on the condition of IR_{LED} = 60mA and 25±2℃ (Room temperature).
- Note 4: Each LED light bar consists of 52 pcs LED package (4 strings x 13 pcs / string).

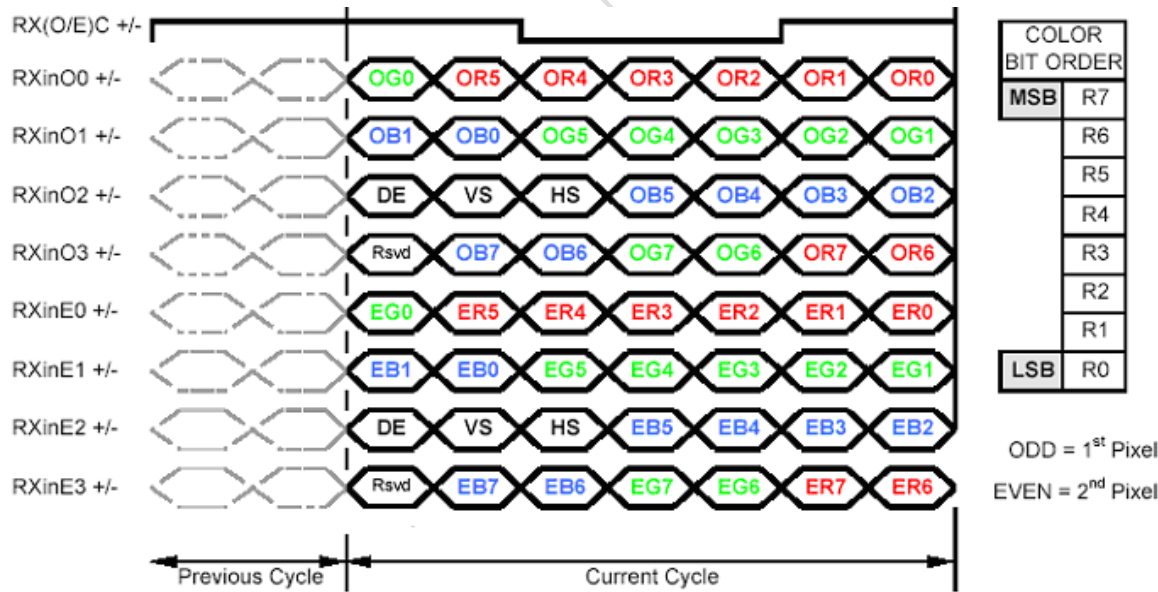
6. Signal Characteristic

6.1 Pixel Format Image

Following figure shows the relationship of the input signals and LCD pixel format.



6.2 The Input Data Format



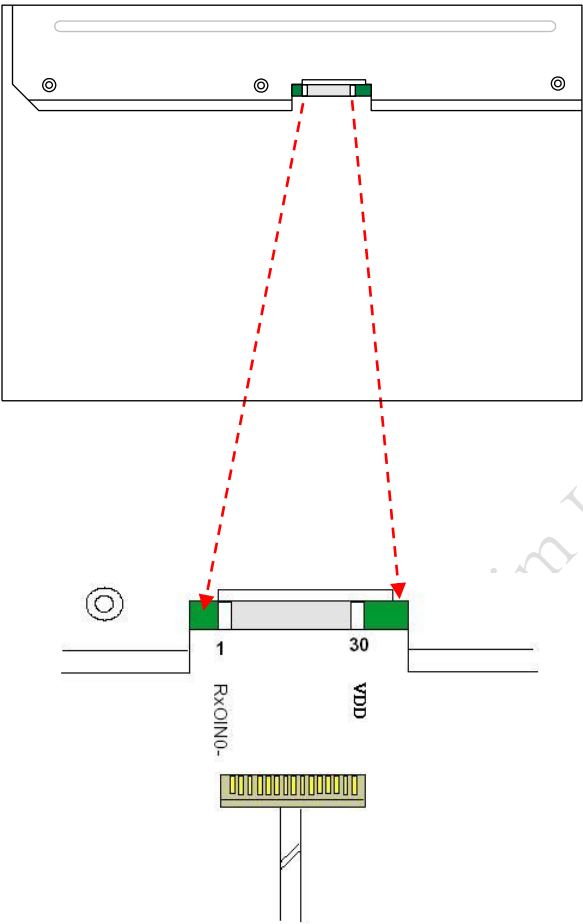
Note1: Normally, DE, VS, HS on EVEN channel are not used.
 Note2: Please follow PSWG.
 Note3: 8-bit in

6.3 Signal Description

The module using one LVDS receiver SN75LVDS82(Texas Instruments). LVDS is a differential signal technology for LCD interface and high speed data transfer device. LVDS transmitters shall be SN75LVDS83(negative edge sampling). The first LVDS port(RxOxxx) transmits odd pixels while the second LVDS port(RxExxx) transmits even pixels.

PIN #	SIGNAL NAME	DESCRIPTION
1	RxOIN0-	Negative LVDS differential data input (Odd data)
2	RxOIN0+	Positive LVDS differential data input (Odd data)
3	RxOIN1-	Negative LVDS differential data input (Odd data)
4	RxOIN1+	Positive LVDS differential data input (Odd data)
5	RxOIN2-	Negative LVDS differential data input (Odd data, DSPTMG)
6	RxOIN2+	Positive LVDS differential data input (Odd data, DSPTMG)
7	GND	Power Ground
8	RxOCLKIN-	Negative LVDS differential clock input (Odd clock)
9	RxOCLKIN+	Positive LVDS differential clock input (Odd clock)
10	RxOIN3-	Negative LVDS differential data input (Odd data)
11	RxOIN3+	Positive LVDS differential data input (Odd data)
12	RxEIN0-	Negative LVDS differential data input (Even data)
13	RxEIN0+	Positive LVDS differential data input (Even data)
14	GND	Power Ground
15	RxEIN1-	Positive LVDS differential data input (Even data)
16	RxEIN1+	Negative LVDS differential data input (Even data)
17	GND	Power Ground
18	RxEIN2-	Negative LVDS differential data input (Even data)
19	RxEIN2+	Positive LVDS differential data input (Even data)
20	RxECLKIN-	Negative LVDS differential clock input (Even clock)
21	RxECLKIN+	Positive LVDS differential clock input (Even clock)
22	RxEIN3-	Negative LVDS differential data input (Even data)
23	RxEIN3+	Positive LVDS differential data input (Even data)
24	GND	Power Ground
25	GND	Power Ground
26	NC	No connection (for AUO test)
27	NC	No connection (for AUO test)
28	POWER	Power +5V
29	POWER	Power +5V
30	POWER	Power +5V

Note1: Start from left side



Note2: Input signals of odd and even clock shall be the same timing.

6.4 Interface Timing

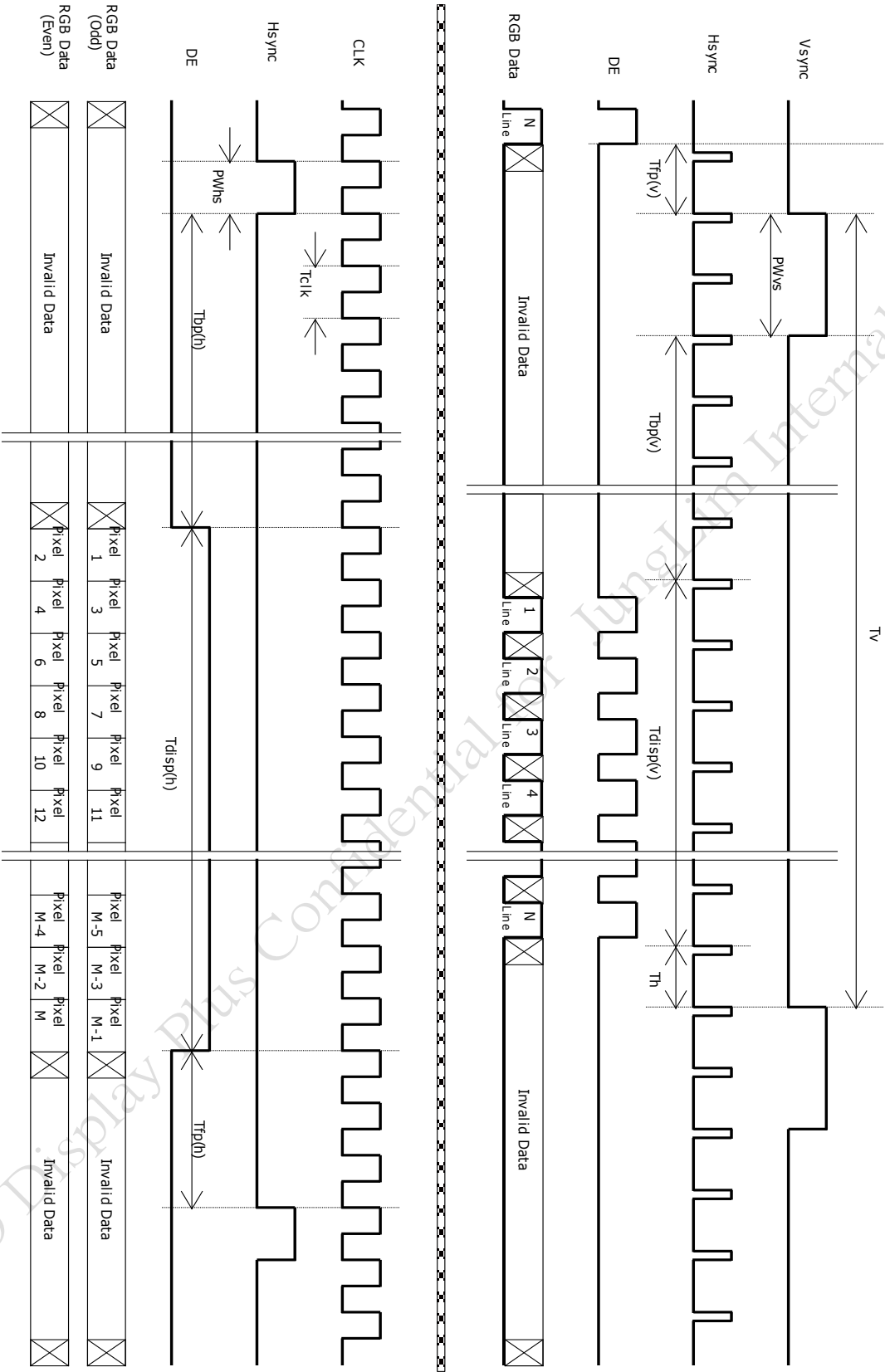
6.4.1 Timing Characteristics

Signal	Item	Symbol	Min	Typ	Max	Unit
Vertical Section	Period	Tv	1032	1066	1150	Th
	Active	Tdisp(v)	1024	1024	1024	Th
	Blanking	Tbp(v)+Tfp(v)+PWvs	8	42	126	Th
Horizontal Section	Period	Th	780	844	871	Tclk
	Active	Tdisp(h)	640	640	640	Tclk
	Blanking	Tbp(h)+Tfp(h)+PWhs	140	204	231	Tclk
Clock	Frequency	Freq.	45	54	67.5	MHz
Frame Rate	Frequency	1/Tv	50	60	75	Hz

Note 1: DE mode only.

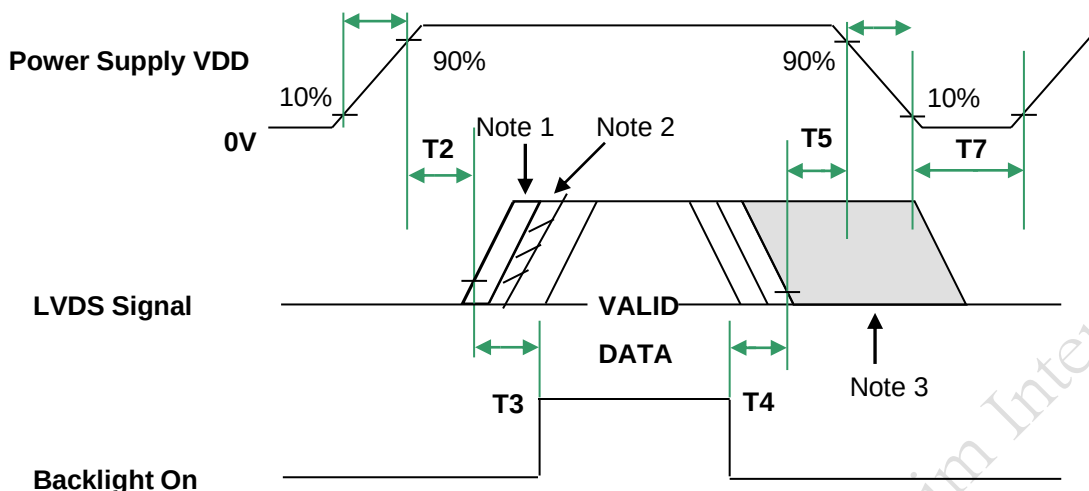
Note 2: Clock Frequency min. < Vertical Period *Horizontal Period * Frame Rate < Clock Frequency max.

6.4.2 Timing Diagram



6.5 Power ON/OFF Sequence

VDD power and lamp on/off sequence are as follows. Interface signals are also shown in the chart. Signals from any system shall be Hi-Z state or low level when VDD is off.



Note1: insert a white pattern 360ms

Note2: insert a black pattern

Note3 :insert a white pattern after valid data and last until VDD falls to 10%.

Note4 :when AC on/off, timing rule of logo power on/off is the same as above.

Power Sequence Timing				
Parameter	Value			Units
	Min.	Typ.	Max.	
T1	0.5	-	10	ms
T2	0	40	50	
T3	500	-	-	
T4	300	-	-	
T5	40	1500	-	
T6	-	-	-	
T7	1000	-	-	

7. Connector & Pin Assignment

Physical interface is described as for the connector on module. These connectors are capable of accommodating the following signals and will be following components.

7.1 TFT LCD Module

7.1.1 Connector

Connector Name / Designation	Interface Connector / Interface card
Manufacturer	P-TWO / JAE
Type Part Number	187034-30091 / FI-XB30SSLA-HF15
Mating Housing Part Number	FI-X30HL FI-X30H (Unlocked Type)

7.1.2 Pin Assignment

Pin#	Signal Name	Pin#	Signal Name
1	RxOIN0-	2	RxOIN0+
3	RxOIN1-	4	RxOIN1+
5	RxOIN2-	6	RxOIN2+
7	GND	8	RxOCLKIN-
9	RxOCLKIN+	10	RxOIN3-
11	RxOIN3+	12	RxEIN0-
13	RxEIN0+	14	GND
15	RxEIN1-	16	RxEIN1+
17	GND	18	RxEIN2-
19	RxEIN2+	20	RxECLKIN-
21	RxECLKIN+	22	RxEIN3-
23	RxEIN3+	24	GND
25	GND	26	NC
27	NC	28	POWER
29	POWER	30	POWER

7.2 Backlight Unit

Physical interface is described as for the connector on module. These connectors are capable of accommodating the following signals and will be following components.

7.2.1 Connector

Connector Name/ Designation	Light Bar Connector
Manufacturer	CviLux
Type Part Number	CI1406M1HRN-NH
Mating Housing Manufacturer	CviLux or compatible
Mating Housing Part Number	CI1406S0000-NH (non-lock type) CI1406SL000-NH (lock type) or compatible

7.2.2 Pin Assignment

Pin no.	Signal name
1	IRLED (current out)
2	IRLED (current out)
3	VLED (voltage in)
4	VLED (voltage in)
5	IRLED (current out)
6	IRLED (current out)

8. Reliability Test

Environment test conditions are listed as following table.

Items	Required Condition	Note
Temperature Humidity Bias (THB)	Ta= 50°C, 80%RH, 300hours	
High Temperature Operation (HTO)	Ta= 50°C, 300hours	
Low Temperature Operation (LTO)	Ta= 0°C, 300hours	
High Temperature Storage (HTS)	Ta= 60°C, 300hours	
Low Temperature Storage (LTS)	Ta= -20°C, 300hours	
Vibration Test (Non-operation)	Acceleration: 1.5 G Wave: Random Frequency: 10 - 200 - 10 Hz Sweep: 30 Minutes each Axis (X, Y, Z)	
Shock Test (Non-operation)	Acceleration: 50 G Wave: Half-sine Active Time: 20 ms Direction: $\pm X$, $\pm Y$, $\pm Z$ (one time for each Axis)	
Drop Test	Height: 60 cm, package test	
Thermal Shock Test (TST)	-20°C/30min, 60°C/30min, 100 cycles	
On/Off Test	On/10sec, Off/10sec, 30,000 cycles	
ESD (Electro Static Discharge)	Contact Discharge: $\pm 8KV$, 150pF(330 Ω) 1sec, 9 points, 25 times/ point.	1
	Air Discharge: $\pm 15KV$, 150pF(330 Ω) 1sec 9 points, 25 times/ point.	
Altitude Test	Operation:10,000 ft Non-Operation:30,000 ft	

Note1: According to EN61000-4-2, ESD class B: Some performance degradation allowed. No data lost

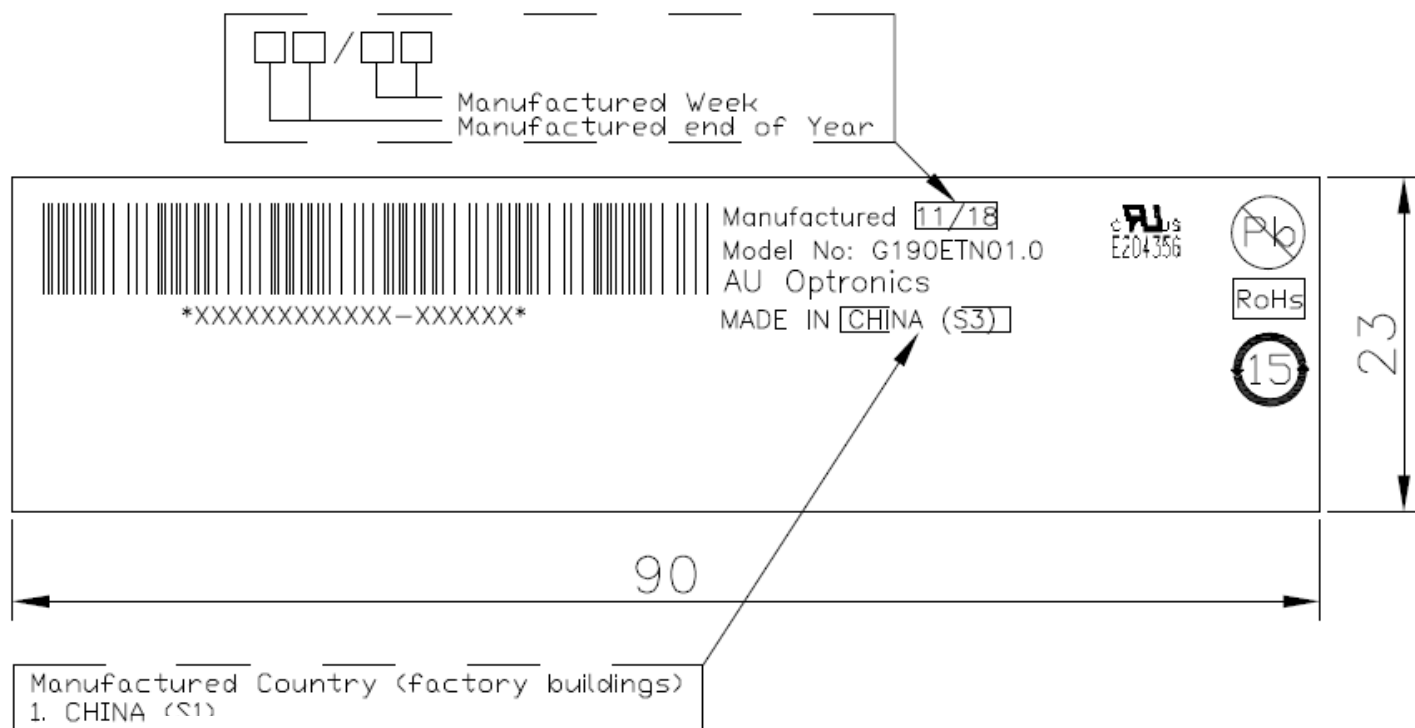
Self-recoverable. No hardware failures.

Note2:

- Water condensation is not allowed for each test items.
- Each test is done by new TFT-LCD module. Don't use the same TFT-LCD module repeatedly for reliability test.
- The reliability test is performed only to examine the TFT-LCD module capability.
- To inspect TFT-LCD module after reliability test, please store it at room temperature and room humidity for 24 hours at least in advance.
- No function failure occurs.

9. Label and Packing

9.1. Shipping Label



9.2 Packing Form

Max. capacity: 13 pieces TFT-LCD module per carton

Max. weight: 26.69 kg per carton

Outside dimension of carton: 409(L)mm* 367(W)mm* 475(H)mm

Pallet : 1150(L)mmx840(W)mmx1082(H)mm

9.3 Palletizing sequence

	pcs / box	box / layer	layer / pallet	pcs / pallet
Shipping by air	13	6	2	156
Shipping by sea	13	6	2	156



AUO Display+

Product Specification

G190ETN01.0

10. Outline Drawing

